Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 22)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 Biomimetics uses knowledge of the adaptations of other species to improve the design of manufactured items.

0 1 . 1 Complete Table 1 by adding the missing adaptation or application.

[4 marks]

Table 1

Species and adaptation	Design application	
Tubercles (bumps) on the flippers of humpback whales		
	Self-cleaning glass	
Shark skin scales that reduce friction		
	Velcro	

0 1 . 2 Explain how the design of termite mounds has been copied to reduce the environmental impact of large buildings.

[3 marks]

Do not write outside the box

0 1 . 3 Describe **two** ways in which studying the physiology of other species could help to solve human health problems.

[2 marks]

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2 .....

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0 1 . 4 Explain why bird wing bones are a useful model for engineers.

[1 mark]

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10

0 2 . 1 Explain how each of the following reduces the environmental impact of fishing.

[8 marks]

Do not write outside the box

Square-mesh escape panels .....

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Minimum and maximum catchable sizes .....

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No-take zones .....

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Closed seasons .....

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0 2 . 2 Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.

[2 marks]

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0 3 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 3 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 3 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

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5

0 4 A student estimated the population of ground beetles in a field using the Lincoln Index. 60 beetles were caught in pitfall traps, marked with a spot of paint and released. Three days later 45 beetles were caught, of which 9 were marked.

0 4 . 1 Calculate the estimated population size.

[2 marks]

Answer ☐ beetles

Do not write outside the box

0 4 . 2 State **four** assumptions of the Lincoln Index method.

[4 marks]

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0 4 . 3 Explain how the estimate would be affected if the paint made the beetles more likely to be eaten by birds.

[2 marks]

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0 4 . 4 Suggest why the second sample was taken after three days rather than the next day or after a month.

[2 marks]

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0 5 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 5 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 5 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

1 .....

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5

0 6 Table 1 shows some features of Earth, Mars and Venus.

Table 1

Feature	Venus	Earth	Mars
Mean distance from Sun / million km	108	150	228
Mass relative to Earth	0.82	1.0	0.11
Atmospheric pressure relative to Earth	92	1.0	0.006
Mean surface temperature / °C	460	15	-60
Magnetic field	Very weak	Strong	Very weak
Rotation period / hours	5 800	24	25

0 6 . 1 Use Table 1 to explain why liquid water cannot exist on the surface of Mars.

[3 marks]

Do not write outside the box

Do not write outside the box

0 6 . 2 Explain how the Earth's mass and its distance from the Sun created suitable conditions for life.

[3 marks]

0 6 . 3 Explain the importance of the Earth's magnetic field and its speed of rotation for living organisms.

[2 marks]

0 6 . 4 Explain how the tilt of the Earth's axis affects living organisms.

[2 marks]

10

0 7 . 1 Define each of the following ecological terms and give an example.

[8 marks]

Species

Ecological niche

Community

Biome

0 7 . 2 Explain the difference between an ecosystem and a biome.

[2 marks]

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0 8 . 1 Describe **four** ways in which the Wildlife and Countryside Act (1981) protects wildlife in the UK.

[4 marks]

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0 8 . 2 Explain the difference between CITES Appendix I and Appendix II, giving an example species for each.

[4 marks]

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0 8 . 3 Explain why controlling international trade can be more effective than trying to stop collection of a species in its own country.

[2 marks]

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0 9 . 1 Explain how each of the following agricultural changes affects wildlife habitats.

[6 marks]

Drainage of wet grassland

Application of fertiliser to a species-rich meadow

Removal of hedgerows to enlarge fields

0 9 . 2 Explain why some farmland habitats are valuable for wildlife conservation.

[2 marks]

0 9 . 3 Describe how agriculture contributes to climate change.

[2 marks]

10

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1 0 . 1 Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

1 0 . 2 Explain why growing coral reefs help to resist climate change.

[2 marks]

1 0 . 3 Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

OR

1 1 . 2 Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS